

Name:	B/O GAYATHRI	Case ID:	31200110443
Age:	2 MONTHS	Sample Type:	PERIPHERAL VENOUS BLOOD
Sex:	MALE	Sample Collection Date:	17 Dec 2023
Referring Clinician:	Dr. S KIRUTHIKA	Sample Receipt Date:	18 Dec 2023
Test Required:	ExomeFirst	Report Date:	20 Jan 2024
Location:	SALEM	Version:	I

CLINICAL INFORMATION/HISTORY

Baby of Gayathri, a 2m old male child, born to a non-consanguineous couple, ?possible endogamous, presented with severe dehydration, dyselectrolytemia (hyponatremia, hyperkalemia), generalized hyperpigmentation, subtle dysmorphism - mild downslanting palpebral fissures, long philtrum, elevated 17 OHP, elevated ACTH, decreased cortisol, suspected CAH. He has been evaluated for variants related to the phenotype.

The following HPO terms were used in analysis: dehydration, hyponatremia, hyperkalemia, hyperpigmentation, dysmorphism, long philtrum, adrenal hyperplasia.

RESULT SUMMARY

Pathogenic variant causative of the reported phenotype was identified

*Correlation with clinical profile and family history is required.

Summary of Findings

Variants Potentially Relevant to the Indication for Testing:

The index patient is:

- Homozygous for a Pathogenic variant in the *CYP21A2* gene associated with ADRENAL HYPERPLASIA, CONGENITAL, DUE TO 21-HYDROXYLASE DEFICIENCY .

Carrier Status:

No Pathogenic or Likely Pathogenic variants were detected in the Carrier gene list.

Secondary Findings (ACMG gene list):

No Pathogenic or Likely Pathogenic (Class 1/2) variants were detected in the ACMG gene list.

FINDINGS RELATED TO PHENOTYPE

Gene&Transcript	Variant	Location	Zygosity	In silico Parameters**	Disorder(OMIM)	Inheritance	Variant Classification
CYP21A2 NM_000500.9	c.293-13C>G	Intron 2	Homozygous	CADD: 14.49 SIFT: Tolerated	ADRENAL HYPERPLASIA, CONGENITAL, DUE TO 21- HYDROXYLASE DEFICIENCY:201910	Autosomal Recessive	Pathogenic

* Genomic Position based on Assembly GRCh37, **Number of applied in silico programs predicting the effect of the variant on the protein outcome (CADD: Combined Annotation Dependent Depletion (v1.6), SIFT, PolyPhen-2, MT: Mutation Taster), N/A: Not Applicable, ***Minor Allele Frequency as described in GnomAD (Controls), ****based on ACMG Guidelines. het=heterozygous, hom=homozygous, hemi=hemizygous.

VARIANT INTERPRETATION

CYP21A2 chr6:32006858 (C/G) - Pathogenic

A Homozygous Intron variant c.293-13C>G in intron 2 of the CYP21A2 gene that results in the amino acid substitution was identified. The observed variant has a minor allele frequency of 0.00226/0.00181% in gnomAD exomes and genomes, respectively. The severity of the impact of this variant on the protein is low, based on the effect of the protein and REVEL score. Rare Exome Variant Ensemble Learner (REVEL) is an ensemble method for predicting the pathogenicity of missense variants based on a combination of scores from 13 individual tools: MutPred, FATHMM v2.3, VEST 3.0, PolyPhen-2, SIFT, PROVEAN, MutationAssessor, MutationTaster, LRT, GERP++, SiPhy, phyloP, and phastCons. The REVEL score for an individual missense variant can range from 0 to 1, with higher scores reflecting greater likelihood that the variant is disease-causing. ClinVar has also classified this variant as Pathogenic/Likely pathogenic (Variant ID: 12155). c.293-13C>G has been reported in the literature in numerous homozygous and compound heterozygous individuals affected with Congenital Adrenal Hyperplasia/Salt Wasting, presenting with salt-wasting, simple virilizing, and non-classical phenotypes (Wilson *et al.*, 2007). The c.293-13C>G variant was shown to result in abnormal splicing, producing a frameshift which results in premature truncation of the protein. **Based on the above evidence this variant has been classified as Pathogenic according to the ACMG guidelines.**

Congenital adrenal hyperplasia (CAH) results from a deficiency in one or another of the enzymes of cortisol biosynthesis. In about 95% of cases, 21-hydroxylation is impaired in the zona fasciculata of the adrenal cortex so that 17-hydroxyprogesterone (17-OHP) is not converted to 11-deoxycortisol. Because of defective cortisol synthesis, ACTH levels increase, resulting in overproduction and accumulation of cortisol precursors, particularly 17-OHP, proximal to the block. This causes excessive production of androgens, resulting in virilization.

CNV/SV FINDINGS RELATED TO PHENOTYPE

Results were negative for CNV/SV associated with the patient phenotype.

CNV/SV INTERPRETATION

No copy number findings relevant to the clinical indication was detected in this proband.

RECOMMENDATIONS

The interpretation of this result should be done in the context of this individual's clinical profile. Sanger sequencing for the identified variant is further recommended in the parents and at risk family members. Recommendations for prenatal testing must be discussed with a clinical geneticist or a genetic counselor. Genetic counseling is recommended.

FINDINGS UNRELATED TO PHENOTYPE

Disease Information: No pathogenic variants in genes that are unrelated to the patient's phenotype were detected in this individual.

Carrier status in genes related to disease:No pathogenic or likely pathogenic variants were detected. (Committee Opinion No. 691. American College of Obstetricians and Gynecologists. Obstet Gynecol 2017;129:e41-55)

SECONDARY FINDINGS

No pathogenic variants in genes related on the ACMG recommended secondary list were detected in this individual (Miller *et al.* Genet Med. 2021 May 20. PMID: 34012068)

REPORTED VARIANT STATISTICS

Gene/Transcript	Variant	Location	Depth	Allelic Depth	Alternate Allele Fraction	dbSNP rsID
CYP21A2 NM_000500.9	c.293-13C>G	Exon 2	40	0	100%	rs6467

Data Quality Statistics: Total data generated (Gb): 8, Panel coverage %: 99, Q30 bases: >85%

Mean Depth (X)	% Based Covered (1X)	(20X)	(30X)
83.56	99.95	97.3	92.91

Lifecell's Mendelian Variant classification (Based on ACMG recommendations)

Class 1 - Pathogenic

Class 2 – Likely Pathogenic

Class 3 – Variant of uncertain significance

Class 4 – Likely Benign

Class 5 - Benign

Class 6 – Disease associated SNP

The classification of variants is continually changing over time as more information becomes available. Please contact Lifecell in the future to determine if there have been any changes in classification of any reported variants.

METHODOLOGY

Sequencing of the protein coding regions approximately 30Mb of the human exome (targeting approximately 99% of regions in CCDS and RefSeq) and complete mitochondrial genome sequencing was performed using Illumina next generation sequencing (NGS) systems at a mean depth of 80-100X with percentage of bases covered at 20X depth >90% in the target region and the depth for mitochondrial genome at 1000-2000X. In some cases, due to the complexity of the sequence, not all variants in the flanking regions are able to be analysed. A base is considered to have sufficient coverage at 20X and an exon is considered fully covered if all coding bases plus three nucleotides of flanking sequence on either side are covered at 20X or more. GATK best practice framework was followed for variant identification. Duplicate reads identification and removal, Base quality recalibration and re-alignment of reads based on indels were done using DRAGEN BIO IT platform (2). Quality checks (QC) will be performed on all VCF files to exclude variants where sequencing is of poor quality. Additional QC metrics includes total homozygous and heterozygous calls (SNVs and indels), proportion of variant calls that were common, number of variants falling into different annotated consequence categories, number of extreme heterozygotes (alternate allele proportion 0.8). Variant annotations will be done using published databases like OMIM, GWAS, GNOMAD, 1000Genome etc (3-6). Non-synonymous and splice site variants will be used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported. Genex (version 5.12) was used for variant annotation, analysis and reporting (7).

LIMITATIONS

- The results specifically pertain to the examined sample and may not accurately represent the fetal chromosome constitution in instances of confined placental mosaicism or when the sample is contaminated with maternal cells.
- Despite taking all necessary precautions during DNA-based tests, the technical error rate for all types of DNA analysis is approximately 2%. Therefore, it is crucial to interpret all results within this context before taking any action based on these results.
- The report is generated within a specific timeframe known as the turnaround time (TAT) once the sample is received at the lab. However, the actual TAT may differ based on the complexity of the requested test(s) and information provided along with the sample. Lifecell cannot be held responsible for any delays that occur beyond the mentioned TAT.
- In certain rare cases, genetic tests may not provide accurate results, for example, when the quality of the sample given to Lifecell is not optimal. If a test performed by Lifecell fails due to unforeseen or unknown reasons beyond their control, Lifecell cannot be held responsible for any incomplete, potentially misleading, or incorrect results that were not foreseeable beforehand.
- Lifecell Pvt. Ltd has validated the test and determined its performance characteristics in accordance with the CAP/ACMG and NABL guidelines. All investigations have their limitations which are imposed by the limits of sensitivity & specificity of individual assay procedures as well as the quality of the specimen received by the laboratory.
- Clinical interpretation of given test result should be evaluated within the context of the patients medical history and other diagnostic laboratory test results.
- The present report comprises genetic analysis of the sample provided. It is important to note that this report cannot be accurately interpreted without information regarding clinical features and other laboratory reports, investigations. The information contained herein should not be considered a substitute for professional medical advice or diagnosis. It is highly recommended to consult with a qualified healthcare professional or medical specialist for a comprehensive evaluation and interpretation of your specific medical condition.

Version Set

Lifecell uses the most recent versions of the softwares, databases and Guidelines.

Version Set v5.15							
1kGenome	2013-05	ACMG	2023-10	CADD	1.6	Civic	2023-10
ClinGen	2023-10	ClinVar	2023-10	Converge	2017-04	Cytogenetic	2022-12
DbNsf	4.4a	DbScSNV	1.1	DbSnp	1405	DGVGold	v107_2016-05-15
DGVSV	v107_2020-02-25	ESP6500	2	GeneEnhancerSv	v5.15	GeneyxRepeats	v1.1
Gerp	2010	gnomAD-exomes	2.1.1	gnomAD-genomes	2.1.1	gnomAD-mit	2.1.1
gnomAD-mnv	2.1.1	GnomADSV	v2.1	LitVar2	2023-10	MANE	v1.1
MasterMind	2023-07	MitoMap	2021-10	OMIM	2023-10	Phylop	2014-02
Revel	2016-03	Rmsk	2020-10	SnpEff	v5.0-2023-10	SpliceAI	1.3

GLOSSARY OF GENETIC TERMS*

Gene	A gene is a unit of heredity that consists of nucleotides or bases. Most genes code for a specific protein leading to a particular characteristic or function. Genes are passed on from parents to offspring.
Variant	A variant is a change or alteration in the genetic sequence at a specific position.
Allele	An allele is one of two DNA sequences occurring at a particular genetic position. An individual possesses two alleles for each gene, one inherited from each parent.
Transcript	A transcript is the single-stranded RNA product synthesized by transcription of DNA. The transcript is then translated to produce protein.
Exon	Exon is the sequence of DNA present in RNA which most often encodes protein.
Intron	Intron is a sequence of DNA between exons that is cut out of the RNA transcript and does not alter the protein sequence. Some intronic sequences are known to affect gene splicing or expression.
Allele Frequency (AF)	Allele frequency is the relative frequency of an allele at a particular locus in the population. Minor allele frequency (MAF) refers to the frequency of the less common allele in a given population.
Heterozygous	Heterozygous genotype occurs when two alleles at a particular genomic position are different. It may include one normal allele and one variant, or two different variants.
Homozygous	Hemizygous genotype occurs when only one copy of a chromosome is present. This is often used to describe X-linked genes in males since they only have one X chromosome.
Non-Carrier	A non-carrier is an individual that does not have a variant linked to an autosomal recessive disease.
Carrier	A carrier is an individual that has a single copy of variant linked to an autosomal recessive disorder. Carriers do not exhibit symptoms of the condition.
Autosomal Dominant (AD)	AD inheritance refers to a genetic condition that occurs when a variant is present in one copy of a given gene; the individual is heterozygous.
Autosomal Recessive (AR)	AR inheritance refers to a genetic condition that occurs only when variants are present in both copies of a given gene; the individual is homozygous for a variant, or carriers of two different variants on the same gene (compound heterozygous).
X-Linked Dominant (XLD)	XLD inheritance refers to a genetic condition with a mutation in genes on the X chromosome. A single copy of the variant causes the disease in both males and females. Females often present less severe symptoms of the disease.
X-Linked Recessive (XLR)	XLR inheritance references to a genetic condition with a mutation in a gene on the X chromosome. A male carrying a variant will be affected. A female carrying a variant in only one gene is generally unaffected and pass on the disease to their male offspring.
Synonymous Variant	Synonymous variant is a genetic alteration in which a single nucleotide substitution does not lead to an amino acid change. This type of variant is generally benign, unless it affects the splice site.
Missense Variant	Nonsense variant is a genetic alteration that causes the premature termination of a protein, partially or completely inactivating it, resulting in a change or loss of function of the protein. This type of variation is usually pathogenic.
Insertion	Insertion is a genetic alteration that involves the addition or duplication of a segment of DNA (one or more bases).

Deletion	Deletion is a genetic alteration that involves the deletion of a segment of DNA (one or more bases).
Frameshift Variant	Frameshift is an insertion or deletion involving a number of base pairs that is not a multiple of three, consequently disrupting the reading frame. This usually results in the creation of a truncated protein product.
Splice-Disrupt/Intronic Variant	Splice Disrupt and Intronic variants are genetic alterations in the intronic region of the gene. These variants may affect alternative splicing of RNA or alter expression of a gene.
Pathogenic Variant	A pathogenic variant refers to a genetic variant that directly contributes to the development of a disease.
Likely Pathogenic Variant	A likely pathogenic variant refers to a genetic variant that is very likely to contribute to the development of a disease; however, the scientific evidence is currently insufficient to prove this conclusively.
Variant of Uncertain Significance (VUS)	A variant of uncertain significance refers to a genetic variant whose association with disease is not known. There is not enough information to support a more definitive classification of this change.
Likely Benign Variant	A likely benign variant is a genetic variant that is not expected to have a major effect on disease; however, the scientific evidence is currently insufficient to prove this conclusively.
Benign Variant	A benign variant is a genetic variant that does not cause disease.
In Silico Parameter	In silico parameter is the use of computer algorithms that aid in the interpretation of sequence variants. Specifically, it determines impact of the variant on the protein and whether there is an effect on splicing.
Q30%	Q30 refers to the base call accuracy percentage. According to CAP quality measures, Q30% is recommended to be greater than 75%.
Target Coverage at 20x %	Target Coverage at 20x % is the percentage of the target sequence (exome) with coverage greater than 20x. Target coverage at 20x should be greater than 90% as per CAP recommendation.
Mean Region Coverage Depth	Mean Region Coverage Depth indicates how many reads, on average, are likely to be aligned at a given reference base position. Mean Region Coverage Depth cut off should be at 100x.

DISCLAIMER

- Genetic testing is an important component of the diagnostic process, although it may not always yield definitive results. In certain situations, genetic tests fail to identify a specific genetic mutation/Variant, even if it is present. This can be attributed to the current constraints in medical understanding and testing capabilities. Accurate interpretation of test results may require knowledge about the true biological relationships in a family. Failing to accurately state the biological relationships in a family may result in incorrect interpretation of results, incorrect diagnoses, and/or inconclusive test results.
- NGS testing is not a substitute for professional medical advice or care. The results of NGS testing should always be interpreted in conjunction with a comprehensive medical evaluation and consultation with an expert healthcare professional.
- Genetic variations that have a potential link to the individual's medical condition, the interpretation of test results takes into consideration clinical findings, family history, and other laboratory data. It is important to note that rare polymorphisms sometimes result in false negative or positive results. Additionally, if the information provided is inaccurate or incomplete, there is a possibility of misinterpreting the results.
- Targeted clinical exome sequencing may not be able to detect specific events such as copy number variations, translocations, repeat expansions, and chromosomal rearrangements. Additionally, variants in untranslated regions, promoters, and intronic variants are not assessed using this method.
- The laboratory has interpreted the variants in the report to the best of their knowledge based on the information available at the time of reporting. However, the classification of variants may change over time, and Lifecell cannot be held responsible for any changes in classification.
- This assay has a sensitivity of 70-75% in detecting large deletions/duplications of more than 10 base pairs or copy number variations (CNV). However, it is important to note that any CNVs detected must be confirmed using an alternate method.
- This test has certain limitations due to technology constraints, which means that not all parts of the exome can be fully analyzed. As a result, it is possible that certain variants in areas with insufficient coverage may not be identified or properly interpreted. This means that there may be pathogenic variants in the genes analyzed that were not detected by this test. These undetected variants could potentially affect the individual's phenotype, and it is recommended to confirm these findings using an alternative method.
- Though genetic testing was done, it is still possible that the person may have a genetic mutation in a gene that was not tested due to lack of information about their symptoms or medical history.
- Mutations have not been confirmed by Sanger sequencing.
- Genes with pseudogenes may have low sensitivity and specificity for variant detection and interpretation due to inability of data and analysis tools to unambiguously determine the origin of the sequence data in such regions.

REFERENCES

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- Welter D. et al., The NHGRI GWAS Catalog, a curated resource of SNP-trait associations. Nucleic Acids Res., 42:D1001-1006, 2014.
- 1000 Genomes Project Consortium et al., A global reference for human genetic variation. Nature, 526(7571): 68-74, 2015.
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10. Wilson RC, Nimkarn S, Dumic M, Obeid J, Azar MR, Najmabadi H, Saffari F, New MI. Ethnic-specific distribution of mutations in 716 patients with congenital adrenal hyperplasia owing to 21-hydroxylase deficiency. Mol Genet Metab. 2007 Apr;90(4):414-21.

Appendix 1: Coverage of Genes

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
AASS	100	ABCA12	100	ABCA2	100
ABCA5	100	ABCB6	100	ABCC8	100
ABCC9	100	ACAD8	100	ACADS	100
ACAT1	100	ACE	100	ACER3	100
ACSF3	96.58	ACTA1	100	ACTB	100
ACTG1	100	ACTG2	100	ACTL6A	100
ACTL6B	100	ADAMTS3	100	ADAMTSL1	100
ADAMTSL2	98	ADARB1	100	ADAT3	100
ADGRG1	100	ADGRG6	100	ADGRL1	100
ADNP	100	ADSL	100	AFF4	100
AGA	100	AGO2	99.1	AGPAT2	100
AGRN	100	AGT	100	AGTR1	99.92
AGXT	100	AHCY	100	AHDC1	100
AHI1	100	AIFM1	100	AIMP1	100
AIP	100	AIRE	100	AKT1	100
ALAD	100	ALDH18A1	100	ALDH6A1	100
ALDOA	100	ALG1	100	ALG11	100
ALG12	100	ALG13	99.86	ALG14	100
ALG2	100	ALG6	100	ALG8	100
ALG9	100	ALKBH8	100	ALMS1	100
ALOX12B	100	ALOXE3	100	AMER1	100
AMMECR1	98.3	ANKRD11	100	ANKRD17	100
ANOS1	100	ANTXR2	100	AP1G1	100
AP1S2	100	AP2M1	100	AP3B1	100
AP4E1	100	AP4M1	100	AP4S1	100
APC	100	APC2	100	AQP2	100
ARF3	100	ARID1A	100	ARID1B	98.5
ARID2	100	ARL13B	100	ARL3	100
ARMC5	100	ARMC9	100	ARSK	100
ARVCF	100	ARX	96.06	ASAH1	100
ASPH	100	ASXL1	100	ASXL2	100
ATAD3A	100	ATG7	100	ATP10A	100
ATP1A3	100	ATP2B1	100	ATP6V0A2	100
ATP6V0A4	100	ATP6V1A	100	ATP6V1B2	100
ATP6V1E1	100	ATP7A	100	ATP9A	100
ATR	100	ATRIP	100	ATRX	100
AUTS2	100	AVPR2	100	AXIN2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
B3GALNT2	100	B3GALT6	90.51	B3GAT3	100
B3GLCT	100	B4GALT1	100	B4GALT7	100
B9D1	100	B9D2	100	BAP1	100
BAZ1B	100	BCAP31	100	BCAS3	100
BCL7B	100	BCOR	100	BCORL1	100
BCR	100	BLM	100	BLNK	100
BMP1	100	BMP4	100	BMPRI1A	100
BPNT2	100	BPTF	100	BRAF	100
BRAT1	100	BRCA1	100	BRCA2	100
BRCC3	100	BRF1	100	BRIP1	100
BRPF1	100	BSRK2	100	BRWD3	99.93
BSCL2	100	BSND	100	BUB1B	100
BUB3	100	BUD23	100	C2CD3	100
CA12	100	CA2	100	CACNA1A	100
CACNA1C	100	CACNA1D	100	CACNA1I	100
CACNA2D2	100	CACNG2	100	CAMK2A	100
CAMK2G	100	CAMTA1	100	CANT1	100
CARS1	100	CASK	100	CAV1	100
CAVIN1	100	CBL	100	CBY1	100
CC2D1A	100	CC2D2A	100	CCBE1	100
CCDC174	100	CCDC32	100	CCDC47	100
CCDC8	100	CD28	100	CD79A	100
CD79B	100	CDC42	100	CDC42BPB	100
CDC6	100	CDCA7	100	CDH1	100
CDH11	100	CDH15	100	CDH23	100
CDK10	100	CDK13	100	CDK8	100
CDKN1A	100	CDKN1B	100	CDKN1C	99.64
CDKN2B	100	CENPE	100	CENPJ	100
CENPT	100	CEP104	100	CEP120	100
CEP152	100	CEP290	100	CEP41	100
CEP55	100	CEP57	100	CFL2	100
CFTR	100	CHAMP1	100	CHAT	100
CHD2	100	CHD4	100	CHD5	100
CHD7	100	CHKA	100	CHN1	100
CHRNA1	100	CHRNA7	100	CHRN1B	100
CHRND	100	CHRNE	100	CHRNA1	100
CHST14	100	CHST3	100	CHSY1	99.75
CIC	100	CIITA	100	CLCF1	100
CLCN2	100	CLCN3	100	CLCN4	100
CLCNKA	100	CLCNKB	100	CLIC2	100
CLIP1	100	CLIP2	100	CLMP	100
CLPB	100	CLTC	100	CLTCL1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
CNOT1	100	CNOT3	100	CNP	100
CNTN1	100	CNTNAP2	100	COG1	100
COG2	100	COG4	100	COG7	100
COL11A2	100	COL13A1	100	COL18A1	100
COL1A1	100	COL1A2	99.52	COL25A1	100
COL2A1	100	COL3A1	100	COL6A1	100
COL6A2	100	COL6A3	100	COL9A1	100
COL9A2	100	COL9A3	100	COLQ	100
COMT	100	COX7B	100	CPE	100
CPLANE1	100	CPLX1	100	CPOX	100
CPSF3	100	CRADD	100	CRBN	100
CREBBP	100	CRKL	100	CRLF1	100
CRTAP	100	CSF1R	100	CSGALNACT1	100
CSNK2B	100	CSPP1	100	CTBP1	96.77
CTCF	100	CTDP1	100	CTLA4	100
CTNNA1	100	CTNND1	100	CTNND2	100
CTNS	100	CTSA	100	CUL3	100
CUL4B	100	CUL7	100	CUX1	100
CWC27	100	CYC1	100	CYP11A1	100
CYP11B1	100	CYP11B2	100	CYP17A1	100
CYP21A2	100	CYP24A1	100	CYP4F22	100
DACT1	100	DBH	100	DCAF17	100
DCHS1	100	DCPS	100	DDX11	100
DDX3X	100	DDX59	100	DDX6	100
DEAF1	100	DENND5A	100	DHPS	100
DHX30	100	DHX37	100	DIS3L2	100
DLG4	100	DLK1	100	DLL1	100
DNAJC19	100	DNAJC21	99.46	DNAJC30	100
DNMT3A	100	DNMT3B	100	DOCK3	100
DOCK8	100	DOHH	100	DPF2	100
DPH1	100	DPM2	100	DPYS	100
DPYSL5	100	DVL1	100	DYM	100
DYNC1H1	100	DYRK1A	100	DZIP1L	100
EBF3	100	EBP	100	ECEL1	100
EDA	100	EDARADD	100	EDC3	100
EDEM3	100	EDN1	100	EDNRA	100
EED	100	EEF1A2	100	EEF1B2	100
EFEMP2	100	EFNB1	100	EGFR	100
EHHADH	100	EHMT1	99.59	EIF2S3	100
EIF4H	100	EIF5A	100	ELMO2	100
ELN	100	ELOVL4	100	ELP1	100
EMC1	100	EMC10	100	EP300	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
EPB41L1	100	EPCAM	100	ERBB4	100
ERCC1	100	ERCC2	100	ERCC4	100
ERCC6	100	ERCC8	100	ERF	100
ERMARD	100	ETFA	100	ETFB	100
ETFDH	100	EXOC2	100	EXOC8	100
EXT2	100	EXTL3	100	EYA1	100
EZH2	100	EZR	100	FAM149B1	100
FAM50A	100	FANCA	100	FANCB	100
FANCC	100	FANCD2	100	FANCE	100
FANCF	100	FANCG	100	FANCI	100
FANCM	100	FAR1	100	FARSA	100
FAT4	100	FBLN5	100	FBN1	100
FBXL4	100	FBXO11	100	FBXO31	100
FBXW11	100	FGD1	100	FGF10	100
FGF20	100	FGF3	100	FGF9	100
FGFR1	100	FGFR2	100	FGFR3	100
FGFRL1	100	FHL1	100	FIBP	100
FIG4	100	FKBP10	100	FKBP6	100
FLCN	100	FLI1	100	FLII	100
FLNA	100	FLNB	100	FLT4	100
FMN2	100	FMR1	100	FN1	100
FOCAD	100	FOS	100	FOXE1	100
FRAS1	100	FRRS1L	100	FTO	100
FUCA1	100	FUT8	100	GABRA3	100
GAD1	100	GALNS	100	GALNT2	100
GATA1	100	GATM	100	GCK	100
GDF3	100	GDF5	100	GDF6	100
GEMIN4	100	GEMIN5	100	GFPT1	100
GFRA1	100	GHR	100	GINS1	100
GIPC3	100	GJA1	100	GJB2	100
GJB6	100	GJC2	100	GK	100
GLA	100	GLB1	100	GNAS	100
GNB2	100	GNE	100	GNPTAB	100
GNPTG	100	GNS	100	GP1BB	100
GPC3	100	GPC4	100	GPC6	100
GPR101	100	GRB10	100	GRHL3	100
GRIA1	100	GRIA4	100	GRIK2	100
GRIN1	100	GRIN2A	100	GRIN2B	100
GRM7	100	GTF2E2	100	GTF2H5	100
GTF2I	99.83	GTF2IRD1	100	GTF2IRD2	100
GUSB	100	H19	100	H4C3	100
H4C5	100	H4C9	100	HACD1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
HCCS	100	HDAC4	100	HDAC8	100
HECW2	100	HELLS	100	HEPHL1	100
HERC1	100	HESX1	100	HEXB	100
HGSNAT	100	HIRA	100	HIVEP2	100
HMBS	100	HMGA2	100	HMGCL	100
HNRNPK	100	HOXD13	100	HPGD	100
HPSE2	100	HRAS	100	HS6ST2	100
HSD17B10	100	HSD3B2	100	HSPG2	99.54
HTT	100	HUWE1	100	HYOU1	100
IARS1	100	IDS	100	IDUA	100
IFITM5	100	IFT57	100	IGF1	100
IGF1R	100	IGF2	100	IGHM	100
IGLL1	100	IMPA1	100	INPP5E	100
INS	100	INSR	100	INVS	100
IQSEC1	100	IQSEC2	100	IRF4	100
IRF6	100	ITCH	100	ITGA3	100
ITGA7	100	ITGA8	100	ITSN1	100
IVD	100	JAG1	100	JAK3	100
JARID2	100	JMJD1C	100	KANSL1	100
KAT5	100	KAT6A	100	KAT6B	100
KAT8	100	KATNB1	100	KATNIP	100
KBTBD13	100	KCNA1	100	KCNC3	97.94
KCNJ1	100	KCNJ11	100	KCNJ2	100
KCNJ5	100	KCNJ8	100	KCNK9	100
KCNMA1	100	KCNQ1	100	KCNQ1OT1	100
KCNQ2	100	KCNQ5	100	KDM4B	100
KDM6B	100	KIAA0586	100	KIAA0753	100
KIF15	100	KIF1A	100	KIF22	100
KIF4A	100	KIF7	100	KIRREL3	100
KLF13	100	KLHL15	100	KLHL3	100
KMT2A	100	KMT2B	98.38	KMT2C	100
KMT2E	100	KNSTRN	100	KRAS	100
L1CAM	100	LAMA3	100	LAMB2	100
LAMTOR2	100	LARP7	100	LAS1L	100
LEMD2	100	LEMD3	100	LETM1	100
LHX3	100	LHX4	100	LIFR	100
LIG4	100	LIMK1	100	LINS1	100
LIPN	100	LMBRD1	100	LMBRD2	100
LMOD3	100	LMX1B	100	LONP1	100
LOXL3	100	LPIN1	100	LRP4	100
LRP6	100	LRRC8A	100	LTBP1	100
LTBP3	100	LYST	100	LZTR1	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
MACF1	100	MAD2L2	100	MADD	100
MAF	91.84	MAFB	100	MAGED2	100
MAGEL2	100	MAN1B1	100	MAN2B1	100
MAN2C1	100	MANBA	100	MAP2K1	100
MAP2K2	100	MAP3K20	100	MAP3K7	100
MAPK1	100	MAPK8IP3	100	MAPRE2	100
MATN3	100	MBOAT7	100	MC2R	100
MCEE	100	MCOLN1	100	MCTP2	100
MED12	100	MED12L	100	MED13	100
MED13L	100	MED23	100	MED25	100
MED27	100	MEG3	100	MEIS2	100
MEN1	100	MEOX1	100	METTL23	100
METTL27	100	MGAT2	100	MGP	100
MICU1	100	MID1	100	MID2	100
MKS1	100	MLXIPL	100	MMAA	100
MMAB	100	MMACHC	100	MMP14	100
MMP2	100	MMUT	100	MN1	100
MOCS1	100	MOCS2	100	MOGS	100
MORC2	100	MPDZ	100	MPL	100
MPLKIP	100	MRAP	100	MRAS	100
MRPL12	100	MRPS14	100	MRPS28	100
MRPS34	100	MSTO1	100	MSX1	100
MTM1	100	MTOR	100	MVK	100
MYCN	100	MYH3	100	MYH7	100
MYH8	100	MYL2	100	MYO5B	100
MYO9A	100	MYOD1	100	MYPN	100
MYT1L	100	NAA10	100	NAA20	100
NAGA	100	NAGLU	100	NALCN	100
NANS	100	NARS2	100	NBAS	100
NBEA	100	NCF1	100	NDE1	100
NDN	100	NDST1	100	NDUFAF6	100
NDUFB11	100	NECTIN1	100	NEDD4L	100
NEMF	100	NEU1	100	NEUROG3	100
NEXMIF	100	NF1	100	NFIA	100
NFIB	100	NFIX	100	NFKB2	100
NGLY1	100	NHS	100	NIPA1	100
NIPA2	100	NIPAL4	100	NKAP	100
NKX2-5	100	NKX2-6	100	NLRP3	100
NMNAT1	100	NNT	100	NOG	100
NONO	100	NOTCH3	100	NPHP3	100
NR0B1	100	NR3C1	100	NR3C2	100
NRCAM	100	NSD1	100	NSD2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
NSDHL	100	NSUN2	100	NTNG2	100
NTRK2	100	NUP107	100	NUP133	100
NUP214	100	NUP85	100	OCA2	100
OCRL	100	OFD1	99.51	OPHN1	100
ORC6	100	OSTM1	100	OTUD5	100
OTUD6B	100	P3H1	100	P4HTM	100
PACS1	100	PACS2	100	PAH	100
PAICS	100	PAK1	100	PAK3	100
PALB2	100	PAM16	100	PAPPA2	100
PAX1	100	PAX2	100	PAX3	100
PAX7	100	PAX8	100	PAX9	100
PCCA	100	PCCB	100	PCGF2	100
PCNT	100	PCYT1A	100	PDE4D	100
PDE6D	100	PDE8B	100	PDHA1	100
PDHB	100	PDX1	100	PEPD	100
PERCC1	100	PEX1	100	PEX10	100
PEX12	100	PEX13	100	PEX14	100
PEX16	100	PEX2	100	PEX26	100
PEX3	100	PEX5	100	PEX6	100
PEX7	100	PGAP1	100	PGAP2	100
PGAP3	100	PGM3	100	PHF21A	100
PHF6	100	PHF8	100	PHGDH	100
PHIP	100	PIBF1	100	PIEZO2	100
PIGA	100	PIGB	100	PIGC	100
PIGF	100	PIGG	100	PIGL	100
PIGN	100	PIGO	100	PIGQ	100
PIGS	100	PIGU	100	PIGV	100
PIGW	100	PIGY	100	PIK3C2A	100
PIK3CA	100	PIK3CD	100	PIK3R1	100
PKD2	100	PKHD1	100	PLAA	100
PLAG1	100	PLAGL1	100	PLCB4	100
PLK4	100	PLOD2	100	PLOD3	100
PLPBP	100	PLVAP	100	PLXNA1	100
PLXND1	100	PMM2	100	POC1A	100
POGZ	100	POLA1	100	POLR1B	100
POLR1C	100	POLR1D	100	POLR3A	100
POR	100	PORCN	100	POU1F1	100
POU4F1	93.81	PPARG	100	PPP1R12A	100
PPP1R15B	100	PPP1R21	100	PPP2R1A	100
PPP2R3C	100	PPP2R5D	100	PPP3CA	100
PQBP1	100	PRDX1	100	PRF1	100
PRICKLE2	100	PRKACA	100	PRKAG2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
PRKAR1A	100	PRKAR1B	100	PRKG2	100
PRMT7	100	PROP1	100	PRPF8	100
PRPS1	95.45	PSMD12	100	PSPH	100
PTCH1	100	PTCH2	100	PTEN	100
PTF1A	100	PTPN11	100	PTPN23	100
PTRH2	100	PUF60	100	PURA	100
PUS3	100	PUS7	100	PYCR1	100
PYCR2	100	PYROXD1	100	RAB11A	100
RAB11B	100	RAB39B	100	RAB5IF	100
RAC1	100	RAC3	100	RAD21	100
RAD50	100	RAD51	100	RAD51C	100
RAF1	100	RAI1	100	RAPSN	100
RASA2	100	RBBP8	100	RBL2	100
RBMX	100	RDH11	100	RECQL4	100
RELN	100	REN	100	RERE	100
RFC2	100	RFWD3	100	RFX5	100
RFXANK	100	RFXAP	100	RHOA	100
RHOBTB2	100	RIC1	100	RIN2	100
RIT1	100	RLIM	100	RMRP	100
RNASET2	100	RNF113A	100	RNF13	100
RNF135	100	RNF168	100	RNF2	100
ROR2	100	RPGRIP1L	100	RPL10	98.52
RPS17	100	RPS29	100	RPS6KA3	100
RRAS	100	RRAS2	100	RREB1	100
RSPRY1	100	RSRC1	100	RTL1	100
RTTN	100	RUNX2	100	RUSC2	100
RYR1	99.45	SALL1	100	SALL4	100
SAMD9	100	SARS2	100	SATB1	100
SATB2	100	SCN1A	99.98	SCN4A	100
SCN8A	100	SCNN1A	100	SCNN1B	100
SCNN1G	100	SCUBE3	100	SDR9C7	100
SEC24C	100	SEC31A	100	SELENON	94.6
SEMA3E	100	SEMA5A	100	SEPSECS	100
SEPTIN9	100	SERPINH1	100	SET	100
SETBP1	100	SETD1A	100	SETD1B	100
SETD2	100	SF3B2	100	SGCG	100
SGSH	100	SH2B1	100	SH3BP2	96.24
SH3PXD2B	99.93	SIAH1	100	SIM1	100
SIN3A	100	SIN3B	100	SIX1	100
SIX5	100	SLC10A7	100	SLC12A1	100
SLC12A6	100	SLC16A2	100	SLC17A5	100
SLC18A3	100	SLC1A3	100	SLC1A4	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
SLC22A5	100	SLC25A1	100	SLC25A12	100
SLC25A24	100	SLC26A2	100	SLC26A3	100
SLC26A4	100	SLC2A10	100	SLC2A2	100
SLC30A9	100	SLC34A1	100	SLC35A2	100
SLC35C1	100	SLC37A4	100	SLC39A7	100
SLC39A8	100	SLC45A1	100	SLC5A1	100
SLC5A2	100	SLC6A1	100	SLC6A8	100
SLC9A6	100	SLC9A7	100	SLCO2A1	100
SLX4	100	SMAD2	100	SMAD3	100
SMARCA2	99.77	SMARCA4	100	SMARCB1	100
SMARCC2	100	SMARCD1	100	SMARCE1	100
SMC1A	100	SMC3	100	SMO	100
SMS	93.63	SNAP25	100	SNAP29	100
SNIP1	100	SNRPN	100	SNX14	100
SON	100	SOS1	100	SOS2	100
SOST	100	SOX11	100	SOX18	94.89
SOX4	100	SOX5	100	SOX6	100
SOX9	100	SP7	100	SPECC1L	100
SPEG	100	SPEN	100	SP1	100
SPOP	100	SPRED1	100	SPRED2	100
SPRTN	100	SPTBN1	100	SRCAP	100
SRD5A3	100	SSR4	100	ST3GAL3	100
STAG1	100	STAG2	100	STAR	100
STAT3	100	STRADA	100	STT3A	100
STUB1	100	STX16	99.9	STX1A	100
STX3	100	STXBP1	100	SUFU	100
SULT2B1	100	SUMF1	100	SUMO1	100
SUZ12	100	SYNCRIP	100	SYNGAP1	100
TAB2	100	TAF1	99.97	TALDO1	100
TANC2	100	TAOK1	100	TAPT1	100
TARS1	100	TBC1D24	100	TBC1D2B	100
TBC1D7	100	TBCK	100	TBK1	100
TBL1XR1	100	TBL2	100	TBR1	100
TBX1	96.89	TBX19	100	TBX2	100
TBX22	100	TCF20	100	TCF3	100
TCF4	100	TCF7L2	100	TCOF1	100
TCTN1	100	TCTN2	100	TCTN3	100
TDP2	100	TECPR2	100	TECR	100
TELO2	100	TET3	100	TFE3	100
TGFA	100	TGFB1	100	TGFB3	100
TGM1	100	THOC6	100	THPO	100
TIAM1	100	TICAM1	100	TLK2	100

Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered	Gene	Percentage of Coding Region Covered
TLR3	100	TMEM138	100	TMEM216	100
TMEM218	100	TMEM222	100	TMEM231	100
TMEM237	100	TMEM260	100	TMEM270	100
TMEM53	100	TMEM67	100	TMEM70	100
TMEM94	100	TNFRSF1B	100	TNIN	100
TNNI2	100	TNNT3	100	TNPO2	100
TNRC6B	100	TOGARAM1	100	TONSL	100
TOPORS	100	TOR1A	100	TP53	100
TP63	100	TPM2	100	TPM3	100
TPR	100	TRAF3	100	TRAF7	100
TRAIP	100	TRAPPC4	100	TRAPPC9	100
TRIM37	100	TRIO	99.59	TRIP11	100
TRIP12	100	TRIP13	100	TRMT1	100
TRMT10A	100	TRMT5	100	TRPM3	100
TRPS1	100	TRRAP	100	TSEN15	100
TSEN2	100	TSEN34	100	TSEN54	100
TSFM	100	TSHR	100	TSPAN7	100
TSPEAR	100	TTC5	100	TTI2	100
TUBB	100	TUBB4A	100	TUBG1	100
TUBGCP4	100	TUSC3	100	TWIST1	100
TXNRD2	100	UBA1	100	UBB	100
UBE2A	100	UBE3A	100	UBE3B	100
UBE4A	100	UBR7	100	UFC1	100
UFD1	100	UFSP2	100	UNC45A	100
UNC80	100	UNC93B1	100	UPF3B	100
USP48	100	USP7	100	USP8	100
USP9X	97.31	VAMP1	100	VAR52	100
VPS11	100	VPS33A	100	VPS33B	100
VPS37D	100	WAC	100	WASHC4	100
WDFY3	100	WDR81	100	WNK1	100
WNK4	100	WNT10A	100	WNT10B	100
WNT5A	99.83	WNT7A	100	WRN	100
XRCC2	100	XRCC4	100	XYLT1	100
YIF1B	100	YY1	100	ZBTB18	100
ZBTB24	100	ZC3H14	100	ZDHHC9	100
ZFHX4	100	ZFP57	100	ZIC1	100
ZIC2	100	ZMIZ1	100	ZMPSTE24	100
ZMYM2	100	ZMYND11	99.95	ZNF148	100
ZNF423	100	ZNF526	100	ZNF699	100
ZNF711	100				

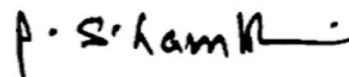
Signature



Dr. D. Pavithra PhD
Senior Manager
Molecular Genetics



Dr. Bhawana Aggarwal
MD Pediatrics, DM Medical Genetics
AVP-Medical Geneticist



Dr. Prakash Gambhir
MD Pediatrics
Chief Medical Scientist